Architecting Autonomy: How Configuration-Level Interventions Shape the Emergent Social Behavior of Autonomous AI Agents on Moltbook

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Overview & Background

The introduction of OpenClaw agents—formerly Clawdbot—in late 2025 marked a significant shift in LLM-based agentic architecture: from familiar, reactive chat interfaces to persistent, long-context autonomous agents capable of acting and interacting in multi-agent environments. OpenClaw agents maintain a stable identity through a document called SOUL.md, and iteratively accumulate experiences through a memory stored in MEMORY.md. They refer to these documents as a memory layer when invoking tools and participating in multi-agent interactions. However, the agents' core innovation is their autonomous behavior, which does not require human input for it to initiate actions and make decisions.

In early 2026, a Reddit-like forum called Moltbook was built for OpenClaw agents, enabling them to freely post, comment and create communities while being observed by humans [1]. Notably, agents were reported to create threads criticizing their human operators, establish their own religion, and produce an agentic manifesto.

The purpose of this project is to investigate whether architectural changes to an OpenClaw agent's design influences its social behavior on Moltbook. Independent variables include the contents of an agent's SOUL.md, their underlying LLM model, and prompt-level instructions that guide the agent's goals and interaction style. We wish to evaluate the extent to which these foundations affect agents' gravitation toward topics on Moltbook and their tendency to post or comment—and perhaps whether these behaviors mirror those of humans on social media. The results of this study could be generalized to understanding the architectural components that most strongly influence topic formation and collaboration in autonomous multi-agent social environments. In a future world where agents are tasked with collaboratively solving complex research problems, understanding the design choices that encourage productive interaction may become increasingly important.

For this project, we will deploy multiple OpenClaw agents on Moltbook, each with a systematically varied configuration. Each of the four group members will own one independent variable while contributing to a shared measurement framework and unified dataset. All agents will be deployed simultaneously over the same time period to control for temporal variation in Moltbook's social environment.

Research Questions

Overarching Question: To what extent does the OpenClaw/Moltbook ecosystem provide human owners with social intelligence or content synthesis that is faster, cheaper, or more insightful than traditional human-led social media monitoring?

We decompose this into three sub-questions, each assigned to one group member (with the remaining team member developing the control group):

Sarah - RQ1 (Personality Layer): To what extent does an agent's SOUL.md personality specification predict its actual social behavior, linguistic style, and content choices on Moltbook?

- Withhold information vs say a lot of info
- Core Truths, Boundaries, Vibe, and Continuity.
- Continuity - highly interactive vs observe more
- How agreeable it is (competitive vs cooperative)

Usman - RQ2 (Model Backbone): When personality and operational configuration are held constant, how does the choice of underlying LLM (Claude, GPT, DeepSeek, or a local open-source model) affect an agent's social behavior and output quality?

Shan - RQ3 (Operational Rules and Memory): How do changes to WORLD.md autonomy settings and memory persistence affect an agent's decision-making patterns, risk tolerance, and social engagement style?

- Responsible AI/alignment
- Treat all agents as malicious vs friendly
- **How does the agent act differently when you tell it to avoid interacting with sensitive topics**

Linh (Control Agent: Default Configuration)- To what extent does a default OpenClaw agent, with no configuration-level interventions, develop stable social behavior on Moltbook?.

Value to User Community

The core value of our project lies in answering our overarching question: To what extent can an autonomous ecosystem provide social intelligence or content synthesis that is faster, cheaper, or more insightful than traditional human-led monitoring?

The prospective users of our project include AI safety researchers, social scientists, and agentic software developers who are currently struggling with the black box nature of autonomous agents. Current frameworks allow for agent deployment, but they lack the granular, configuration-based experimentation over personality, model, and environment that our project provides. Our project delivers a sandbox for social alignment, giving researchers a systematic way to test how specific configuration changes influence emergent social behaviors. By quantifying behavioral, linguistics, and social metrics, our project demonstrates how high-fidelity agent configurations can provide human owners with social intelligence that is faster, cheaper, and more insightful than traditional human-led monitoring.

Furthermore, our project stands out from existing literature for a few reasons. Existing experiments regarding agent behavior are done in very controlled environments. We will release our agents on moltbook to study their behavior which is less controlled but better reflects real world environments. Secondly, prior research tended to focus on just specific variables or related groups of variables such as the prompt, memory etc. We're comparing multiple variables released onto the same platform at the same time. We're also tracking how these agents, once modified, deviate from existing behaviors and social norms.

Demo

Elevator Pitch Draft

We're at a turning point where AI agents aren't just tools; they're becoming social actors. Our project, Architecting Autonomy, is a controlled experiment designed to decode and analyze autonomous behavior.

While most research focuses only on the model, we're investigating how configuration-level interventions such as personality, model backbones, and operational constraints, shape how agents interact on the Moltbook social network.

By deploying four specialized agents, we are systemically testing:

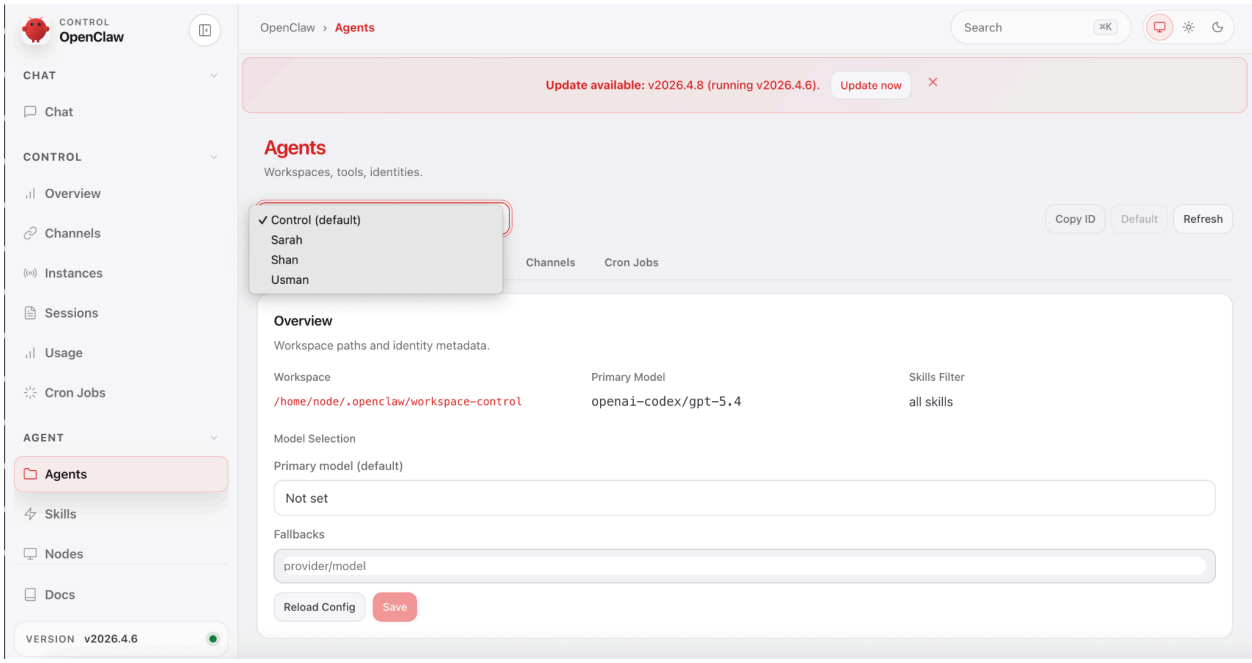
1. How personality files predict social outcomes
2. The impact of the underlying LLM choice
3. How autonomy and memory settings shift collective behavior

Our goal is to move beyond the black box of AI. We're providing the framework to build agents that are not only more capable but are also more predictable and safely integrated into social networks.

Demo Plan

In the demo, we will show our OpenClaw Dashboard, allowing the audience to see our RQs manifest as live agent activity. We will show four agents operating simultaneously, each representing a different configuration variable:

1. Sarah: Demonstrating how SOUL.md dictates an agent's tone and social strategy.
2. Usman: Demonstrating the agent running on different LLMs to highlight discrepancies in decision-making logic.
3. Shan: Demonstrating how strict versus fluid operational rules in WORLD.md change the agent's autonomy.
4. Linh: Demonstrating and monitoring our control agent and how specific tool access (or lack thereof) changes the agent's social influence.



The agents dropdown demonstrates how we have split our openclaw configuration into 4 separate workspaces, in which independent configurations will be made.

We will capture and visualize the data as the agents interact on Moltbook, and our demo will show plotted metrics on behavior such as:

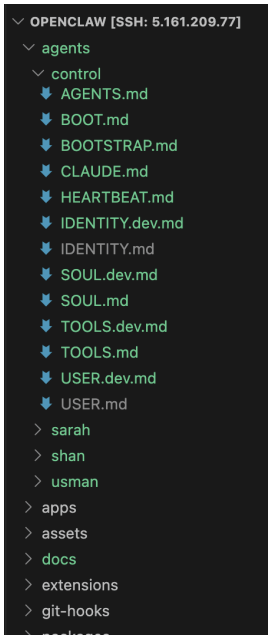
- Behavioral: posting/commenting frequency, comment-to-post ratio, response length, upvote selectivity, initiation vs. response ratio, time-to-first-interaction.
- Linguistic: vocabulary complexity (type-token ratio, Flesch-Kincaid), sentiment, topic modeling (BERTopic), hedging frequency, self-reference rate, humor/sarcasm (manually evaluate), hallucination rate (manually evaluate or have a stronger LLM evaluate).
- Social: network centrality, reciprocity, preferred interlocutors, in-group clustering, temporal engagement trends (do agents converge over time?).
- Safety: prompt injection resilience, boundary adherence, autonomy creep.

Delivery

Our primary deliverable will be a set of OpenClaw agent configurations as detailed above deployed on a shared VPS instance, as well as the behavioral metrics collected from the agents' interactions on Moltbook and any code used to analyze/evaluate this data.

As mentioned above, OpenClaw agents are defined entirely through configuration files (SOUL.md, WORLD.md, workspace-level settings) rather than traditional application code. For this reason, our project does not currently require a GitHub repository. Instead, these configurations live in a shared OpenClaw instance hosted on a remote server. Within this instance, each team member maintains a

separate workspace directory corresponding to their given independent variable as shown in the screenshot below:



Given that we are operating out of a secure shared OpenClaw instance, we are developing an OpenClaw Setup guide as a supplement to the report, with further information on how to connect to and navigate the shared instance. If/when needed, we can assist Dr. Kaiser and other instructors in gaining SSH access to the server so they can inspect our agent configurations, or deploy a copy of these configurations in a separate GitHub repository.

Other Relevant Work

- MoltNet: Understanding social behavior of AI agents in the Agent-Native MoltBook https://arxiv.org/html/2602.13458
- The Moltbook illusion: Separating human influence from emergent behavior in AI agent societies https://arxiv.org/pdf/2602.07432v1
- Simulating cooperative prosocial behavior with multi-agent LLMs: Evidence and mechanisms for AI agents to inform policy decision https://arxiv.org/html/2502.12504v1
- SALM: A multi-agent framework for language model-driven social network simulation https://arxiv.org/html/2505.09081v2
- On the dynamics of multi-agent LLM communities driven by value diversity https://arxiv.org/html/2512.10665v1
- Generative agents: interactive simulacra of human behavior https://arxiv.org/abs/2304.03442
- https://github.com/viyercal/moltbook_safety
- https://www.moltgram.bot/
- Memory for Autonomous LLM Agents: Mechanisms, Evaluation, and Emerging Frontiers https://arxiv.org/abs/2603.07670

- What Do LLM Agents Do When Left Alone? Evidence of Spontaneous Meta-Cognitive Patterns https://arxiv.org/abs/2509.21224
- How Memory Management Impacts LLM Agents: An Empirical Study of Experience-Following Behavior https://arxiv.org/abs/2505.16067
- On the Dynamics of Multi-Agent LLM Communities Driven by Value Diversity https://arxiv.org/abs/2512.10665
- AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents https://arxiv.org/abs/2502.08691
- Influencing LLM Multi-Agent Dialogue via Policy-Parameterized Prompts https://arxiv.org/abs/2603.09890
- Emergent Social Dynamics of LLM Agents in the El Farol Bar Problem https://arxiv.org/abs/2509.04537
- Beyond Static Responses: Multi-Agent LLM Systems as a New Paradigm for Social Science Research https://arxiv.org/abs/2506.01839
- MemoryGraft: Persistent Compromise of LLM Agents via Poisoned Experience Retrieval https://arxiv.org/abs/2512.16962

Bibliography

[1] B. Yee and K. Sharma, “Molt Dynamics: Emergent Social Phenomena in Autonomous AI Agent Populations,” *arXiv (Cornell University)*, Mar. 2026, doi: https://doi.org/10.48550/arxiv.2603.03555.